EE 5239 Nonlinear Optimization

Lecture 2: Gradient Methods III: Analysis

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Announcement

- HW 2 grade points (60 derivation part and 40 Implementation part)
- Book is dense (examples through class + point out important sections)
- Start to think about your project topics; Project proposal Oct. 10th (Temporary)

A General Analysis for Convergence

- Prove convergence to points that satisfy the first order optimality – we call them stationary solutions [figure]
- The direction $\mathbf{d}^r$ cannot be orthogonal to $\nabla f(\mathbf{x}^r)$ [figure]
- **Gradient related condition**: For any sequence $\{\mathbf{x}^r\}$ that converges to a nonstationary point, the corresponding direction $\{\mathbf{d}^r\}$ is bounded and satisfies

$$\lim_{r\to\infty} \langle \nabla f(\mathbf{x}^r), \mathbf{d}^r \rangle < 0$$

- Is this condition satisfied for $\mathbf{d}^r = -\mathbf{D}^r \nabla f(\mathbf{x}^r)$, with $\mathbf{D}^r$ being a positive definite matrix?

A General Analysis of Convergence

- Here we state a few assumptions about the algorithm/problem
- Let $\mathbf{x}^r$ be a sequence generated by a gradient method

$$\mathbf{x}^{r+1} = \mathbf{x}^r + \alpha_r \mathbf{d}^r$$

- $\mathbf{d}^r$ is gradient related
- Here we keep things general by having time-dependent α_r, and general direction $\mathbf{d}^r$ (will specialize later)
- Assume the following Lipschitz continuous condition is satisfied

$$\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \le L\|\mathbf{x} - \mathbf{y}\|, \; \forall \; \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$$

A General Analysis of Convergence (Assumptions)

- Assume either one of the following choices of stepsize
  1. There exits a scalar $\epsilon \in (0,2)$ such that for all r
  $$\epsilon < \alpha_r \leq -\frac{(2-\epsilon)\langle \nabla f(\mathbf{x}^r), \mathbf{d}^r \rangle}{L\|\mathbf{d}^r\|^2}$$
  2. $\alpha_r \to 0$, and $\sum_{r=1}^{\infty} \alpha_r = \infty$ (i.e., $\alpha_r = \frac{1}{r}$)

- **Claim**: $\nabla f(\mathbf{x}^r) \to 0$

- If $\mathbf{d}^r = -\nabla f(\mathbf{x}^r)$, then the first condition becomes
$$\epsilon < \alpha_r \leq \frac{(2-\epsilon)}{L}$$
Therefore, we can pick, for example, $\alpha_r = \frac{1}{L}$

Convergence Analysis

- Given $\mathbf{x}^r$ and the descent direction $\mathbf{d}^r$, the Lipschitz assumption implies (cf. the descent Lemma)

$$f(\mathbf{x}^r + \alpha \mathbf{d}^r) - f(\mathbf{x}^r) \leq \alpha \langle \nabla f(\mathbf{x}^r), \mathbf{d}^r \rangle + \frac{L}{2}\alpha^2 \|\mathbf{d}^r\|^2$$

- Plugging in the upper-bound for α_r:

$$\begin{aligned}
f(x^r + \alpha_r \mathbf{d}^r) - f(\mathbf{x}^r) &\leq \alpha \left(\langle \nabla f(\mathbf{x}^r), \mathbf{d}^r \rangle + \frac{L}{2}\alpha \|\mathbf{d}^r\|^2 \right) \\
&\leq \alpha \left(\langle \nabla f(\mathbf{x}^r), \mathbf{d}^r \rangle - \frac{2-\epsilon}{2} \langle \nabla f(\mathbf{x}^r), \mathbf{d}^r \rangle \right) \\
&= \alpha \underbrace{\frac{\epsilon}{2} \langle \nabla f(\mathbf{x}^r), \mathbf{d}^r \rangle}_{<0} \\
&\leq \frac{\epsilon^2}{2} \langle \nabla f(\mathbf{x}^r), \mathbf{d}^r \rangle \qquad \text{(use the fact } \alpha \geq \epsilon\text{)}
\end{aligned}$$

- Clearly, the objective is always decreasing
- Question: Where have we used the "gradient related" condition?

Convergence Analysis (cont.)

- Assume $\bar{x}$ is a finite nonstationary limit point.
- Then $f(\mathbf{x}^r) \downarrow f(\bar{\mathbf{x}})$
- Then $\langle \nabla f(\mathbf{x}^r), \mathbf{d}^r \rangle \to 0$, because otherwise $f(\bar{\mathbf{x}}) \to -\infty$
- Is this possible? No, the gradient related condition asserts that $\langle \nabla f(\mathbf{x}^r), \mathbf{d}^r \rangle < 0$, and the condition on α_r asserts that $\alpha_r > 0$
- We arrived at a contradiction – the claim is proved
- How about the diminishing stepsize? Same analysis, but much more involved
- Formal Proof: See Proposition 1.2.3 for constant stepsize, Prop. 1.2.1 for Armijo rule.

Diminishing Stepsize

Proposition 1.2.4: (Convergence for a Diminishing Stepsize) Let $\{x^k\}$ be a sequence generated by a gradient method $x^{k+1} = x^k + \alpha^k d^k$. Assume that for some constant $L > 0$, we have

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|, \qquad \forall\, x, y \in \Re^n, \tag{1.26}$$

and that there exist positive scalars c_1, c_2 such that for all k we have

$$c_1\|\nabla f(x^k)\|^2 \leq -\nabla f(x^k)' d^k, \qquad \|d^k\|^2 \leq c_2\|\nabla f(x^k)\|^2. \tag{1.27}$$

Suppose also that

$$\alpha^k \to 0, \qquad \sum_{k=0}^{\infty} \alpha^k = \infty.$$

Then either $f(x^k) \to -\infty$ or else $\{f(x^k)\}$ converges to a finite value and $\nabla f(x^k) \to 0$. Furthermore, every limit point of $\{x^k\}$ is a stationary point of f.

Summary: Convergence Analysis

- **Convergence**: The GD converges to first order optimality ($\nabla f(\mathbf{x}) = 0$)
  1. Basic requirement
  2. Not necessarily convergent to globally optimal
  3. This only says that the algorithm is reasonable (why?)
- Provide an analysis of GD for constant stepsize rule, for any reasonable direction (not necessarily $-\nabla f(\mathbf{x})$)

Outline

1. Applying Gradient Descent to Regression

2. Convergence Rate Analysis

A Simple Example in Regression

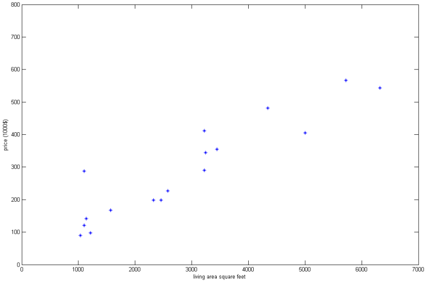
- Let's take a look at a simple example in predicting the house price giving the living areas

Living Area (feet^2)	Price (1000$)
5719	567
3241	345
1139	141
1572	167
1101	287
2576	227
$\vdots$	$\vdots$

- Let's build a regression model and use gradient descent to solve the problem

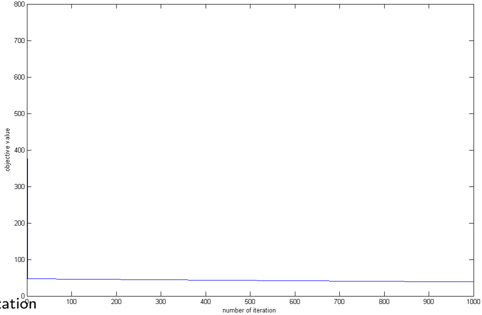
A Simple Example in Regression

- The data is first plotted below (A total of 17 data points)
- Construct the regression model: variables x_1 (slope), x_0 (intercept)
- Data matrix $\mathbf{A} \in \mathbb{R}^{17\times 2}$; first column the area data, second column all 1
- Data vector $\mathbf{b} \in \mathbb{R}^{17\times 1}$, being the housing price.
- Solve the following problem $\min \frac{1}{2}\|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2$



A Simple Example in Regression

- Use the gradient with constant stepsize (how to compute the stepsize?)
- **First Try**: Scale the data so that all the areas are multiplied by 0.01, and all the prices are multiplied by 0.1 – scaling the data
- Run the steepest gradient descent for 1000 iterations; initial $(10, 50)$
- Eigenvalues of $\mathbf{A}'\mathbf{A}$ is 0.0004 and 1.856



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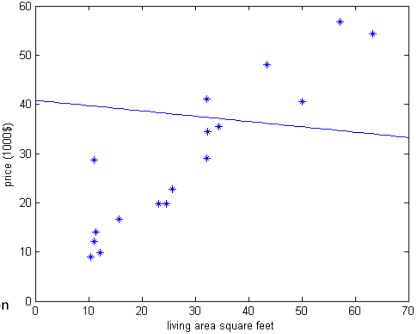

A Simple Example in Regression

- Use the gradient with constant stepsize
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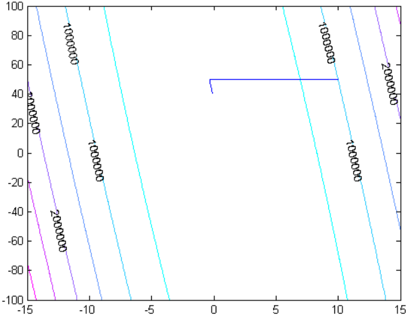

Figure 0.3: The iterates.

A Simple Example in Regression

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- **Second Try**: Scale the data so that all the areas are multiplied by 0.0001, and all the prices are multiplied by 0.1 – scaling the data
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- Eigenvalues of $\mathbf{A}'\mathbf{A}$ is 0.4 and 18.448

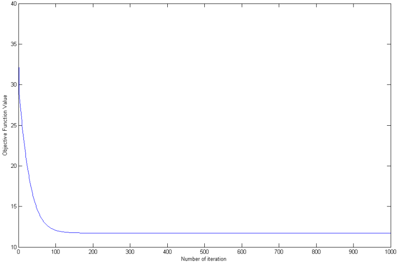

Figure 0.4: The decrease of objective function

A Simple Example in Regression

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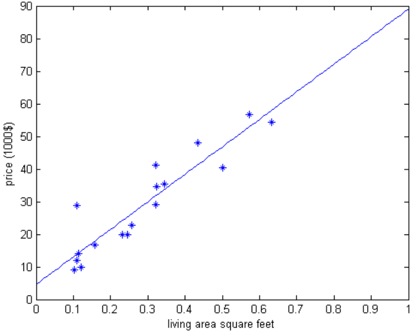

Figure 0.5: The final fitting.

A Simple Example in Regression

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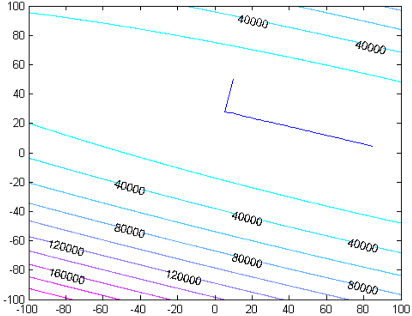


Figure 0.6: The iterates.

Outline

1. Applying Gradient Descent to Regression
2. Convergence Rate Analysis

Convergence Rate Analysis

- Seems that scaling the data gives much different performance on the gradient descent algorithm
- Why? How should we understand such behavior? Practical implications?

Convergence Rate Analysis

- Define an ϵ optimal solution as $\{\mathbf{x}_\epsilon := f(\mathbf{x}^r) - f^* \leq \epsilon\}$
- **Convergence Rate**:
    1. Measures the number of iterations required to get an ϵ optimal solution
    2. Gives global behavior of the algorithm (from start to end)
    3. A popular and important measure for evaluating algorithms in big data related applications
    4. Question: What determines the convergence rate?

Illustration

Figure 0.1: What Determines Convergence Rate?

Illustration

Figure 0.2: What Determines Convergence Rate?

Convergence Rate Analysis

- We focus on a family of special functions, and show that gradient descent methods is able to converge linearly

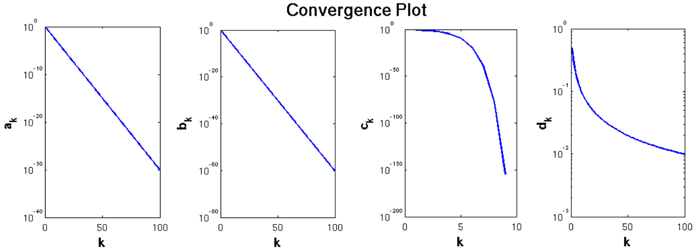
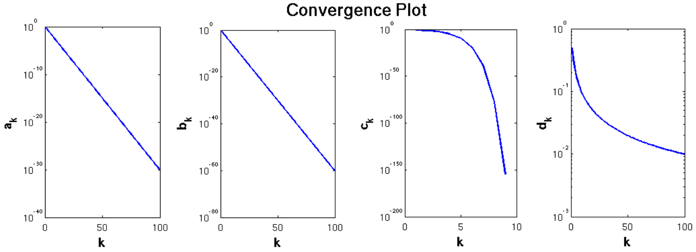


Figure 0.3: Illustration of Convergence Speed. [1] Linear rate (slower); [2] Linear rate (faster);[3] superlinear rate; [4] sublinear rate. y-axis: log of the error, x-axis: iteration number.(Wikipedia: Rate of Convergence)

Convergence Rate Analysis

- We focus on a family of special functions, and show that gradient descent methods is able to converge linearly
- This means some measure of optimality shrinks by a constant factor at each iteration



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- We focus on a family of special functions, and show that gradient descent methods is able to converge linearly
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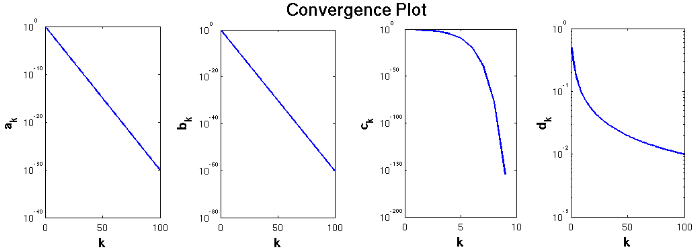
- For example, define the error as $e(\mathbf{x}) := f(\mathbf{x}) - f(\mathbf{x}^*) \geq 0$



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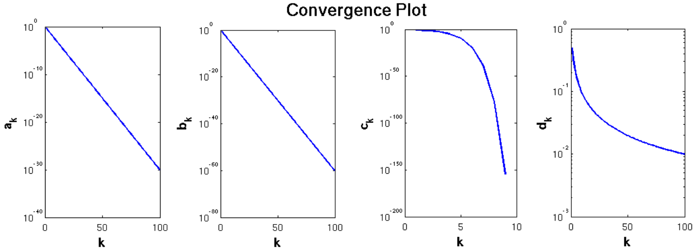
- For example, define the error as $e(\mathbf{x}) := f(\mathbf{x}) - f(\mathbf{x}^*) \geq 0$
- Then,e.g., $e(\mathbf{x}^{r+1}) \leq 0.1 \times e(\mathbf{x}^r)$



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Convergence Rate Analysis

- Linear convergence means that there exists $\beta \in (0,1)$

$$\lim_{r\to\infty} \sup \frac{e(\mathbf{x}^{r+1})}{e(\mathbf{x}^r)} < \beta$$

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- If the above is true for all r, it means $e(\mathbf{x}^{r+1}) \le (\beta)^r e(\mathbf{x}^0)$, or (note $\beta < 1$, so $\ln(\beta) < 0$)

$$\underbrace{\ln(e(\mathbf{x}^{r+1}))}_{\text{log of error}} \le \underbrace{r\ln(\beta)}_{\text{“linear” in iteration \#}} + \ln(e(\mathbf{x}^0))$$

- Log of error a linear function in # iteration r (with negative slope)!

Convergence Rate Analysis

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- Log of error a linear function in # iteration r (with negative slope)!
- Superliner convergence means

$$\lim_{r\to\infty} \sup \frac{e(\mathbf{x}^{r+1})}{e^p(\mathbf{x}^r)} < \beta$$

for some constant $p > 1$

Convergence Rate Analysis

- Why "$\limsup$" is needed?
- This says that as long as in the limit, the linear convergence behavior occurs, we call the algorithm "linearly convergent"
- But we will analyze an algorithm that is much "stronger", in the sense it satisfies (for all iterations)

$$\frac{e(\mathbf{x}^{r+1})}{e(\mathbf{x}^{r})} < \beta, \ \forall \ r$$

with some $\beta \in (0, 1)$.

Strongly Convex Function

- Recall that f is continuously differentiable, f is convex iff (HW 2 problem) [graphically]

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle.$$

- If f is twice continuously differentiable and convex, then

$$f \text{ is convex} \Leftrightarrow \nabla^2 f(\mathbf{x}) \succeq 0, \text{ for all } \mathbf{x}$$

- **A New Notion**: f is strongly convex iff exists $\sigma > 0$

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle + \frac{\sigma}{2} \|\mathbf{x} - \mathbf{y}\|^2.$$

- The entire function has "enough curvature" [graphically]

Strongly Convex (SC) Function

- A function is "strongly convex" if (intuitively)
  1. It is convex
  2. It has no "flat" regions

- If f is twice continuously differentiable, then exists $\sigma > 0$

$$f \text{ is strongly convex} \Leftrightarrow \nabla^2 f(\mathbf{x}) \succeq \sigma \mathbf{I}, \text{ for all } \mathbf{x}$$

- Note, for two matrices, the notation $A \succeq B$ means $A - B \succeq 0$. That is, $A - B$ is a positive semidefinite matrix

- **Example**: A quadratic function $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T \mathbf{A}\mathbf{x}$ with strictly positive definite $\mathbf{A}$, i.e., $\mathbf{A} \succeq \sigma \mathbf{I}$; Here σ is the smallest eigenvalue for $\mathbf{A}$

GD for SC Function: Step 1

- Let's begin analysis of gradient descent for strongly convex problems
- **Goal** To see what does convergence speed depends on

GD for SC Function: Step 1

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- **Goal** To see what does convergence speed depends on
- For simplicity, consider the case where $\mathbf{d}^r = -\nabla f(\mathbf{x}^r)$
- Then $\alpha_r = \frac{1}{L}$ and (from the discussion from the previous lecture)

$$f(\mathbf{x}^{r+1}) = f(\mathbf{x}^r + \alpha_r \mathbf{d}^r) \leq f(\mathbf{x}^r) - \frac{1}{2L}\|\nabla f(\mathbf{x}^r)\|^2$$

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- **Goal** To see what does convergence speed depends on
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$$f(\mathbf{x}^{r+1}) = f(\mathbf{x}^r + \alpha_r \mathbf{d}^r) \le f(\mathbf{x}^r) - \frac{1}{2L}\|\nabla f(\mathbf{x}^r)\|^2$$

- This shows that after one round of algorithm, the objective function achieves "sufficient descent"
- The amount of the descent can be measured by the size of the gradient

GD for SC Function: Step 2

- Using the strong convexity assumption, we have

$$f(\mathbf{x}^*) \geq f(\mathbf{x}^r) + \langle \nabla f(\mathbf{x}^r), \mathbf{x}^* - \mathbf{x}^r \rangle + \frac{\sigma}{2}\|\mathbf{x}^r - \mathbf{x}^*\|^2 \tag{0.1}$$

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- Let us view the right hand side function as a function of x^*

$$g(\mathbf{z}) = f(\mathbf{x}^r) + \langle \nabla f(\mathbf{x}^r), \mathbf{z} - \mathbf{x}^r \rangle + \frac{\sigma}{2}\|\mathbf{x}^r - \mathbf{z}\|^2 \qquad (0.2)$$

GD for SC Function: Step 2

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- Minimizing the right hand side over $\mathbf{z}^*$ we obtain (optimal solution is $z^* = x^r - \frac{1}{\sigma}\nabla f(x^r)$)

$$\begin{aligned} f(\mathbf{x}^*) &\geq f(\mathbf{x}^r) + \langle \nabla f(\mathbf{x}^r), \mathbf{z}^* - \mathbf{x}^r \rangle + \frac{\sigma}{2}\|\mathbf{x}^r - \mathbf{z}^*\|^2 \\ &\geq f(\mathbf{x}^r) - \frac{1}{2\sigma}\|\nabla f(\mathbf{x}^r)\|^2 \end{aligned}$$

My current functional value is not too far away from my goal!

GD for SC Function: Step 3

- Step 1 tells us how much descent we have at each step
- Step 2 tells us how close we are to the global min
- Combing the previous two steps, we have

$$
\begin{aligned}
f(\mathbf{x}^{r+1}) - f(x^*) &\overset{\text{Step 1}}{\leq} f(\mathbf{x}^r) - f(\mathbf{x}^*) - \frac{1}{2L}\|\nabla f(\mathbf{x}^r)\|^2 \\
&\overset{\text{Step 2}}{\leq} f(\mathbf{x}^r) - f(\mathbf{x}^*) - \frac{\sigma}{L}(f(\mathbf{x}^r) - f(\mathbf{x}^*))
\end{aligned}
$$

- Rearranging terms, use the definition $e(x) := f(x) - f(x^*)$:

$$e(\mathbf{x}^{r+1}) \leq (1 - \frac{\sigma}{L})e(\mathbf{x}^r) := \beta e(\mathbf{x}^r)$$

GD for SC Function

- Linear convergence, with constant $\beta = (1 - \frac{\sigma}{L}) \in (0, 1)$ (why?)
- L/σ is so-called the condition number of f (recall Lecture 1)
  - σ: the smallest eigenvalue of the Hessian of f (recall our quadratic problem)
  - L: the largest eigenvalue of the Hessian of f (recall our quadratic problem)
- Large condition number implies large β
- L/σ big: ill-conditioned (slow convergence for GD)
- L/σ small: well-conditioned (fast convergence for GD)

Summary of the Proof

- **Summary of proof**: Two steps:
  1. S1: Sufficient Descent (the decrease achieved after each iteration):
$$f(\mathbf{x}^{r+1}) - f(\mathbf{x}^r) \le \ldots$$
  2. S2: Estimating cost-to-go (how far away we are from the optimal):
$$f(\mathbf{x}^r) - f(\mathbf{x}^*) \le \ldots$$

GD for SC Function

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- How many iterations are needed for $e(\mathbf{x}^r)$ to reach below ϵ?
- Suppose $e(\mathbf{x}^0) = D_0$, then $e(\mathbf{x}^r) = \beta^r D_0 \le \epsilon$, so we require:

$$r \ge -\ln(D_0/\epsilon)/\ln(\beta) = \ln\left(\frac{D_0}{\epsilon}\right)/\ln(\frac{1}{\beta})$$

- As long as the total # of iteration is larger than the right hand side above, we are guaranteed to reach an ϵ optimal solution

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- Larger the β, the large the number of iterations required!
- The number of iterations scale with error by: $\ln(1/\epsilon)$

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- Larger the β, the large the number of iterations required!
- The number of iterations scale with error by: $\ln(1/\epsilon)$
- **Remark:** For regular convex problems, sublinear convergence rate $r \ge 1/(\epsilon)$ (to be discussed later)

On condition number

- The condition number being large, or **ill-conditioning**, usually arises when the optimization variables are scaled incoherently
- Example:
  1. x_1: gas flow, lbs/hr, nominal value $11,000$
  2. x_2: waste buildup: lbs, nominal value 6×10^{-4}
  3. x_3: steam thermal resistance, $\texttt{BTU/(hr.}ft^2\texttt{.F)}$, nominal value 100
- Should rescale the units so that the nominal values are balanced
- Algorithmically, use other types of gradient descent methods, choose different scaling matrix $\mathbf{D}^r$

On condition number: Scaling variables

- Specifically, perform "data preprocessing"
- Suppose we have data sets $\{\mathbf{a}_i, b_i\}_{i=1}^{M}$ (M data points), $\mathbf{a}_i \in \mathbb{R}^K$ (K features)
- Arrange it into a data matrix $\{\mathbf{A}, \mathbf{b}\}$, where $\mathbf{A} \in \mathbb{R}^{M \times K}$
  1. For each "feature" (or each column of $\mathbf{A}$), say $\mathbf{A}_k$
  $$\mathbf{A}'_k = \frac{\mathbf{A}_k - \bar{\mathbf{A}}_k}{\sigma_{\mathbf{A}_k}}$$
  where $\bar{\mathbf{A}}_k$ is the sample mean and $\sigma_{\mathbf{A}_k}$ is the sample standard deviation (standardization)
  2. For the $\mathbf{b}$, do: $\mathbf{b}' = \mathbf{b} - \bar{\mathbf{b}}$ (centering)
- Then once we obtain $\mathbf{A}'_k$, we can transform back using

$$\mathbf{A}_k = \sigma_{\mathbf{A}_k} \mathbf{A}'_k + \bar{\mathbf{A}}_k$$

Conclusion

- How to select the constant stepsize for gradient descent (inverse proportional to the curvature)
- How to analyze gradient descent
- Practical performance of GD
- Analyze its convergence rate, and its relationship with the condition number